Data mesh is neither a silver bullet nor a plug-and-play solution. Instead, its successful implementation requires overcoming an array of challenges. This section covers the technological, data management, and organizational challenges.

1.7.1 Technological challenges

The most crucial technological challenge of data mesh implementation is ensuring that proper tooling is available to both domain-focused and central platform data teams. Providing only data discoverability and links between domains may prevent the whole ecosystem from becoming a siloed world.

The other challenge is ensuring that the tooling developed locally can be shared across domains. The flexibility and autonomy of teams should not lead to inefficiencies at the company level and reduced synergy. To effectively develop and maintain their data solutions, domain data teams need safe and secure access to data storage services. Centralizing, for example, maintenance of a cloud computing environment is much more efficient than the multiplication of this effort by each team. In addition, centralized control over cloud storage and computing can also help introduce a proper spending regime. We've seen way too often seemingly rational spending by multiple separated teams sum up to amounts far exceeding levels justifiable at the company level.

With a plethora of distributed, semi-independent data products, the development of consistent monitoring, alerting, and logging procedures becomes a challenge. As a result, both central platform and domain teams need to collaborate closely to ensure continuous insight and control over data quality and security and maintenance efficiency.

1.7.2 Data management challenges

Like the first technological challenge, the critical data-management-related challenge is ensuring that data is easy to locate and comprehensively documented. The importance of a data catalog cannot be overestimated. It is not a challenge unique to data mesh, but as it's based on the decentralization paradigm, the consequences of overlooking it would be much more dire. No central data team will be capable of ad hoc identification of available data